

Networking & Cybersecurity Track
Simulated Employment Program

Lab Report:
Configure Access Control

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Background / Scenario

Authentication and authorization are distinct security processes in the world of identity and access management (IAM). Authentication uses passwords and other identification methods to confirm that users are who they say they are. By contrast, authorization assigns user permissions to the resources that the user is allowed access.

In this Packet Tracer (PT) activity, you will configure authentication and authorization for network services including wireless network access, email, and FTP services.

***Note:** This activity opens in Physical mode. However, you can also complete it in Logical mode.*

***Note:** Most tasks in this activity are graded. Click Check Results at any time to view your correct and incorrect Assessment Items.*

Part 1: Configure and Use AAA Authentication Credentials

Step 1: Configure user accounts on the AAA server.

1. Navigate to Headquarters and then click the Wiring Closet, which is the tall, black server chassis in the bottom left corner.



2. On the right side Rack, click AAA-RADIUS server > Services tab, and then AAA under SERVICES.

AAA-RADIUS

Physical Config **Services** Desktop Programming Attributes

SERVICES

AAA

Service ☐ On ☒ Off Radius Port 1812

Network Configuration

Client Name Client IP

Secret ServerType Radius

	Client Name	Client IP	Server Type	Key	
1	WLC-1	192.168.99.250	Radius	WLC-auth!	Add
					Save
					Remove

User Setup

Username Password

	Username	Password	
1	1stFLprn	LPT1PRN	Add
			Save
			Remove

3. Turn on the AAA service.

AAA-RADIUS

Physical Config **Services** Desktop Programming Attributes

SERVICES

HTTP

DHCP

DHCPv6

TFTP

DNS

SYSLOG

AAA

NTP

EMAIL

FTP

IoT

VM Management

Radius EAP

AAA

Service ☒ On ☐ Off Radius Port 1812

Network Configuration

Client Name Client IP

Secret ServerType Radius

	Client Name	Client IP	Server Type	Key	
1	WLC-1	192.168.99.250	Radius	WLC-auth!	Add
					Save
					Remove

4. Under User Setup, add the following usernames and passwords:

Username	Password
user1	PASSuser1!
user2	PASSuser2!

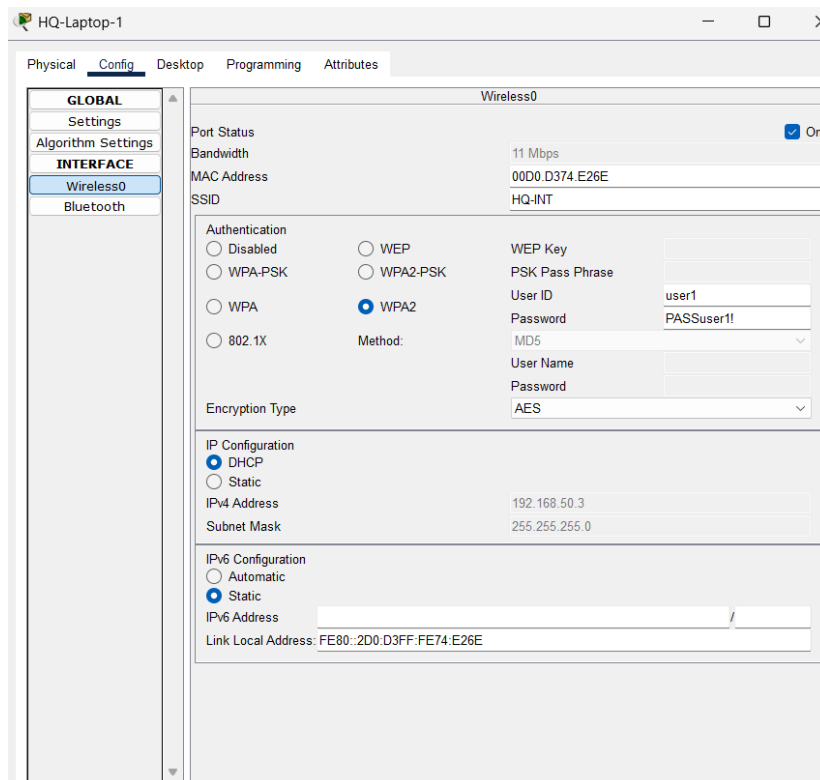
User Setup

Username Password

	Username	Password	
1	1stFLprn	LPT1PRN	Add
2	user1	PASSuser1!	
3	user2	PASSuser2!	
			Save
			Remove

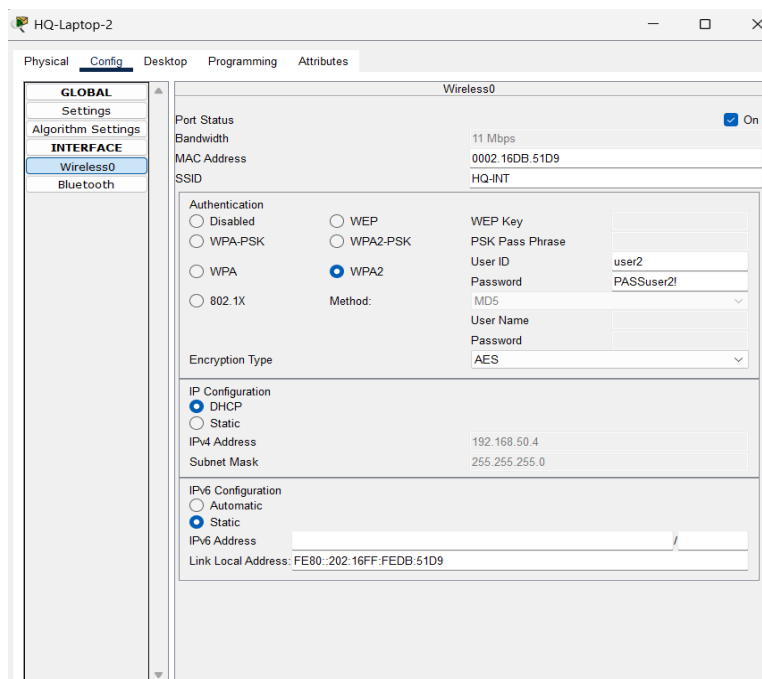
Step 2: Configure wireless authentication on HQ-Laptop-1.

5. Navigate back to HQ and click HQ-Laptop-1. It is located two rooms to the right from the Wiring Closet.
6. Click Config tab, and then under INTERFACE, click Wireless0.
7. In the SSID box, type HQ-INT.
8. In the Authentication area, click WPA2.
9. In the User ID box, enter user1; and enter PASSuser1! in the Password box.
10. In the IP Configuration section, click DHCP. Wait a few moments for the DHCP offer to be accepted.



Step 3: Configure wireless authentication on HQ-Laptop-2.

11. Click HQ-Laptop-2, which is located in the top right corner of HQ.
12. Repeat the previous steps to configure the wireless settings for HQ-Laptop-2, using the user2 credentials.

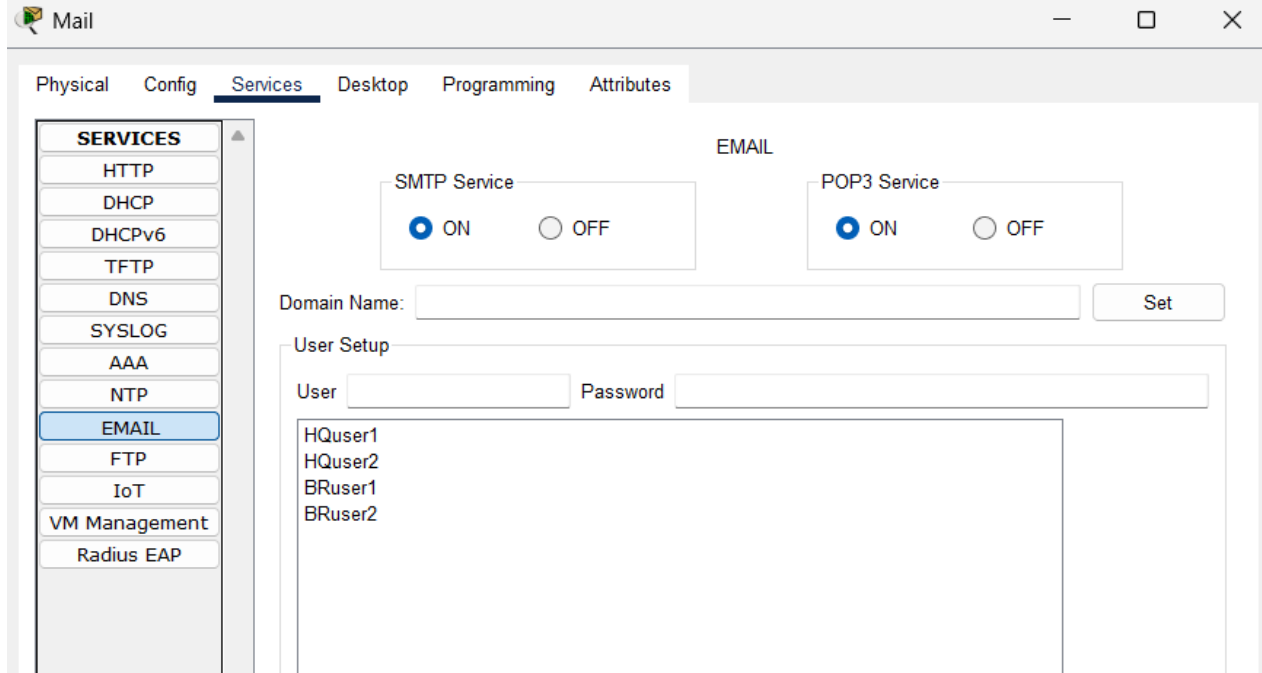


Part 2: Configure and Use Email Services

Step 1: Activate email services and configure email user accounts.

13. Navigate to the Wiring Closet.
14. On the right side Rack, click Mail server > Services tab, and then EMAIL under SERVICES.
15. Turn on both the SMTP and POP3 services.
16. Set the domain to mail.cyberhq.com.
17. Under User Setup, enter the following usernames and passwords. Click the plus sign (+) to add each pair.

Username	Password
HQuser1	Cisco123!
HQuser2	Cisco123~
BRuser1	Cisco123-
BRuser2	Cisco123+



The screenshot shows the 'Mail' configuration window. The 'Services' tab is active, and the 'EMAIL' section is expanded. The 'SMTP Service' and 'POP3 Service' are both set to 'ON'. The 'Domain Name' field is empty, and there is a 'Set' button next to it. Under the 'User Setup' section, a list of users is displayed: HQuser1, HQuser2, BRuser1, and BRuser2.

Step 2: Configure email clients.

18. Navigate back to HQ and click PC 1-1, which is in the bottom corner.
19. Click Desktop tab > Email. The Configure Mail settings open.
20. Enter the following information for PC 1-1 (Suk-Yi):

Your Name	Suk-Yi
Email Address	HQuser1@mail.cyberhq.com
Incoming & Outgoing Mail Server	mail.cyberhq.com
User Name	HQuser1
Password	Cisco123!
Action	Click Save

1-1

Physical
Config
Desktop
Programming
Attributes

Configure Mail
X

User Information

Your Name: Suk-Yi

Email Address HQuser1@mail.cyberhq.com

Server Information

Incoming Mail Server mail.cyberhq.com

Outgoing Mail Server mail.cyberhq.com

Logon Information

User Name: HQuser1

Password:

Save Remove Clear Reset

21. Use the information in the table below to configure email settings for PC 2-3, HQ-Laptop-1, and Net-Admin.

Device	Name	Email Address	Username	Password
PC 1-1	Suk-Yi	HQuser1@mail.cyberhq.com	HQuser1	Cisco123!
PC 2-3	Ajulo	BRuser1@mail.cyberhq.com	BRuser1	Cisco123-
HQ-Laptop-1	Malia	HQuser2@mail.cyberhq.com	HQuser2	Cisco123~
Net-Admin	Net-Admin	BRuser2@mail.cyberhq.com	BRuser2	Cisco123+

2-3

Physical Config **Desktop** Programming Attributes

Configure Mail X

User Information

Your Name: Ajulo

Email Address BRuser1@mail.cyberhq.com

Server Information

Incoming Mail Server mail.cyberhq.com

Outgoing Mail Server mail.cyberhq.com

Logon Information

User Name: BRuser1

Password:

Save Remove Clear Reset

Figure: PC 2-3 — Email Client Configured (Ajulo / BRuser1)

HQ-Laptop-1

Physical Config **Desktop** Programming Attributes

Configure Mail X

User Information

Your Name: Malia

Email Address BRuser2@mail.cyberhq.com

Server Information

Incoming Mail Server mail.cyberhq.com

Outgoing Mail Server mail.cyberhq.com

Logon Information

User Name: BRuser2

Password:

Save Remove Clear Reset

Figure: HQ-Laptop-1 — Email Client Configured (Malia / HQuser2)

The screenshot shows the 'Net-Admin' application window with the 'Desktop' tab selected. A 'Configure Mail' dialog box is open, containing three sections: 'User Information', 'Server Information', and 'Logon Information'. The 'User Information' section has 'Your Name' set to 'Cisco' and 'Email Address' set to 'HQuser2@mail.cyberhq.com'. The 'Server Information' section has both 'Incoming Mail Server' and 'Outgoing Mail Server' set to 'mail.cyberhq.com'. The 'Logon Information' section has 'User Name' set to 'HQuser2' and 'Password' masked with dots. At the bottom of the dialog are buttons for 'Save', 'Remove', 'Clear', and 'Reset'.

Section	Field	Value
User Information	Your Name	Cisco
	Email Address	HQuser2@mail.cyberhq.com
Server Information	Incoming Mail Server	mail.cyberhq.com
	Outgoing Mail Server	mail.cyberhq.com
Logon Information	User Name	HQuser2
	Password	

Figure: Net-Admin — Email Client Configured (BRuser2)

Step 3: Send an email as Suk-Yi.

22. On PC 1-1, click Compose.

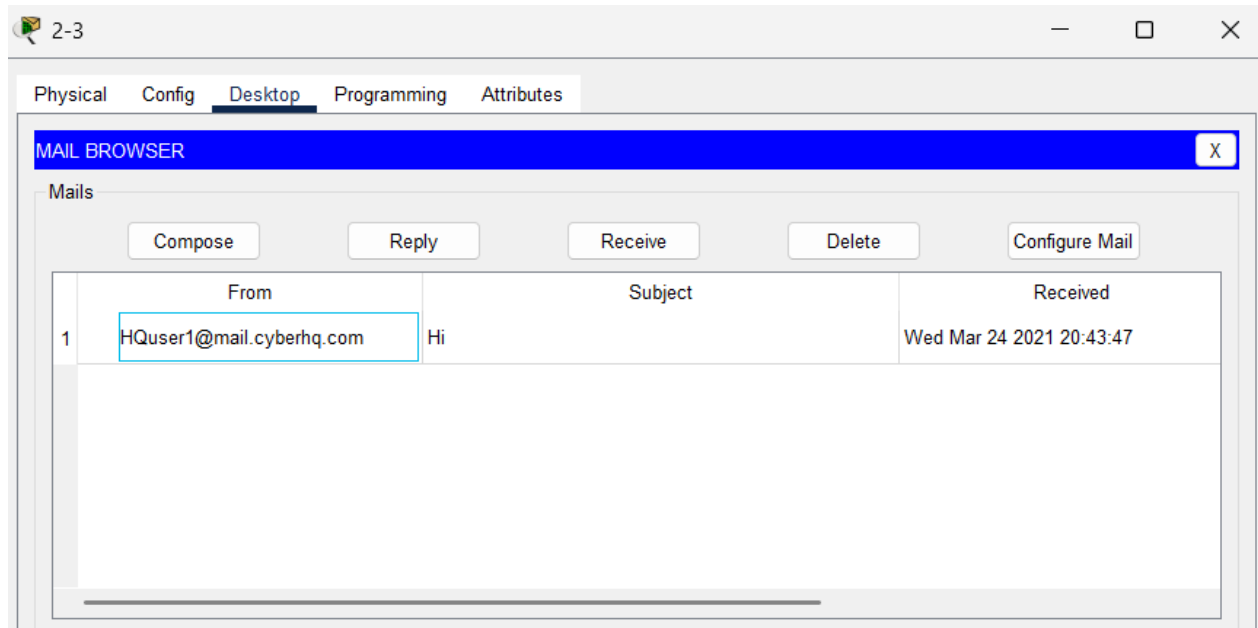
23. Compose an email to Ajulo at BRuser1@mail.cyberhq.com. Enter a subject and email message of your choice. Click Send when finished.

The screenshot shows the 'Net-Admin' application window with the 'Desktop' tab selected. A 'Compose Mail' dialog box is open. It features a 'Send' button, a 'To:' field with 'BRuser1@mail.cyberhq.com', a 'Subject:' field with 'Hi', and a text area containing 'Hello Ajulo'.

Field	Value
To:	BRuser1@mail.cyberhq.com
Subject:	Hi
Message	Hello Ajulo

24. Navigate to Ajulo's PC 2-3. If necessary, click Desktop tab > Email.

25. Click Receive and read the email from Suk-Yi.



Part 3: Configure and Use FTP Services

Step 1: Activate the FTP Service.

26. Navigate to the Wiring Closet.
27. On the right side Rack, click FTP server > Services tab, and then FTP under SERVICES.
28. Turn on the FTP service.

The screenshot shows a web-based configuration interface for an FTP service. The window title is "FTP". At the top, there are tabs: "Physical", "Config", "Services" (which is selected), "Desktop", "Programming", and "Attributes". On the left side, there is a "SERVICES" menu with various options: HTTP, DHCP, DHCPv6, TFTP, DNS, SYSLOG, AAA, NTP, EMAIL, FTP (which is highlighted in blue), IoT, VM Management, and Radius EAP. The main area is titled "FTP" and contains a "Service" toggle switch set to "On". Below this, there is a "User Setup" section with input fields for "Username" and "Password". Underneath these fields are five checkboxes: "Write", "Read", "Delete", "Rename", and "List", all of which are currently unchecked. A table below the checkboxes shows a single user entry with the following data:

	Username	Password	Permission
1	cisco	cisco	RWDNL

To the right of the table are three buttons: "Add", "Save", and "Remove".

Step 2: Create the FTP user accounts.

29. Under User Setup, enter the following usernames, passwords, and privileges. Click Add to add each user.

Note: Be sure username malia does not include Delete as one of the user privileges.

Username	Password	Permissions	Delete?
sukyi	cisco123	Read, Write, Delete, Rename, List	Yes
malia	cisco123	Read, Write, Rename, List	No — NOT included
admin	cisco123	Read, Write, Delete, Rename, List	Yes

Physical Config **Services** Desktop Programming Attributes

SERVICES

- HTTP
- DHCP
- DHCPv6
- TFTP
- DNS
- SYSLOG
- AAA
- NTP
- EMAIL
- FTP**
- IoT
- VM Management
- Radius EAP

FTP

Service ☒ On ☐ Off

User Setup

Username Password

☐ Write ☐ Read ☐ Delete ☐ Rename ☐ List

	Username	Password	Permission
1	cisco	cisco	RWDNL
2	sukui	cisco123	RWDNL
3	ajulo	cisco321	RWDNL
4	malia	cisco123	RWNL

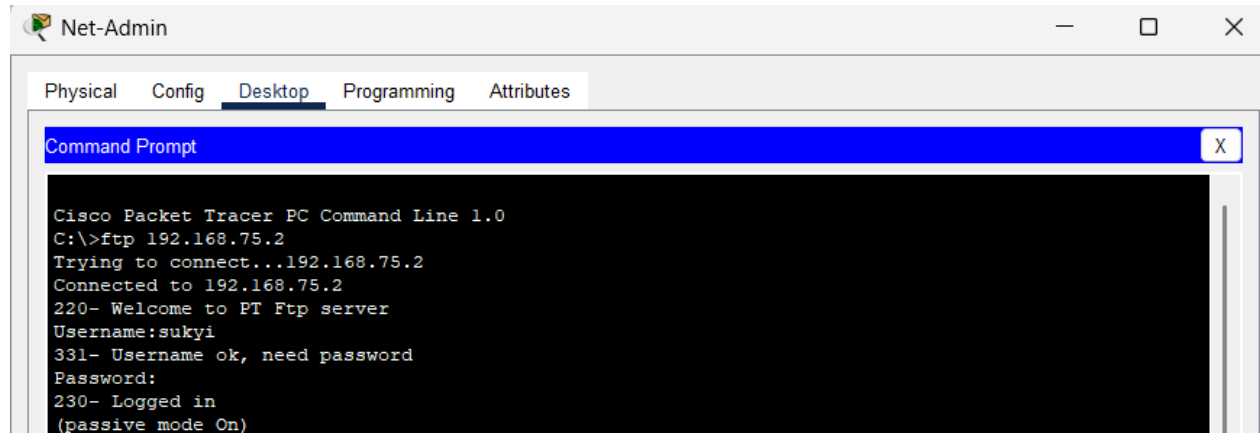
Add Save Remove

30. Verify each user is correctly created and close the server.

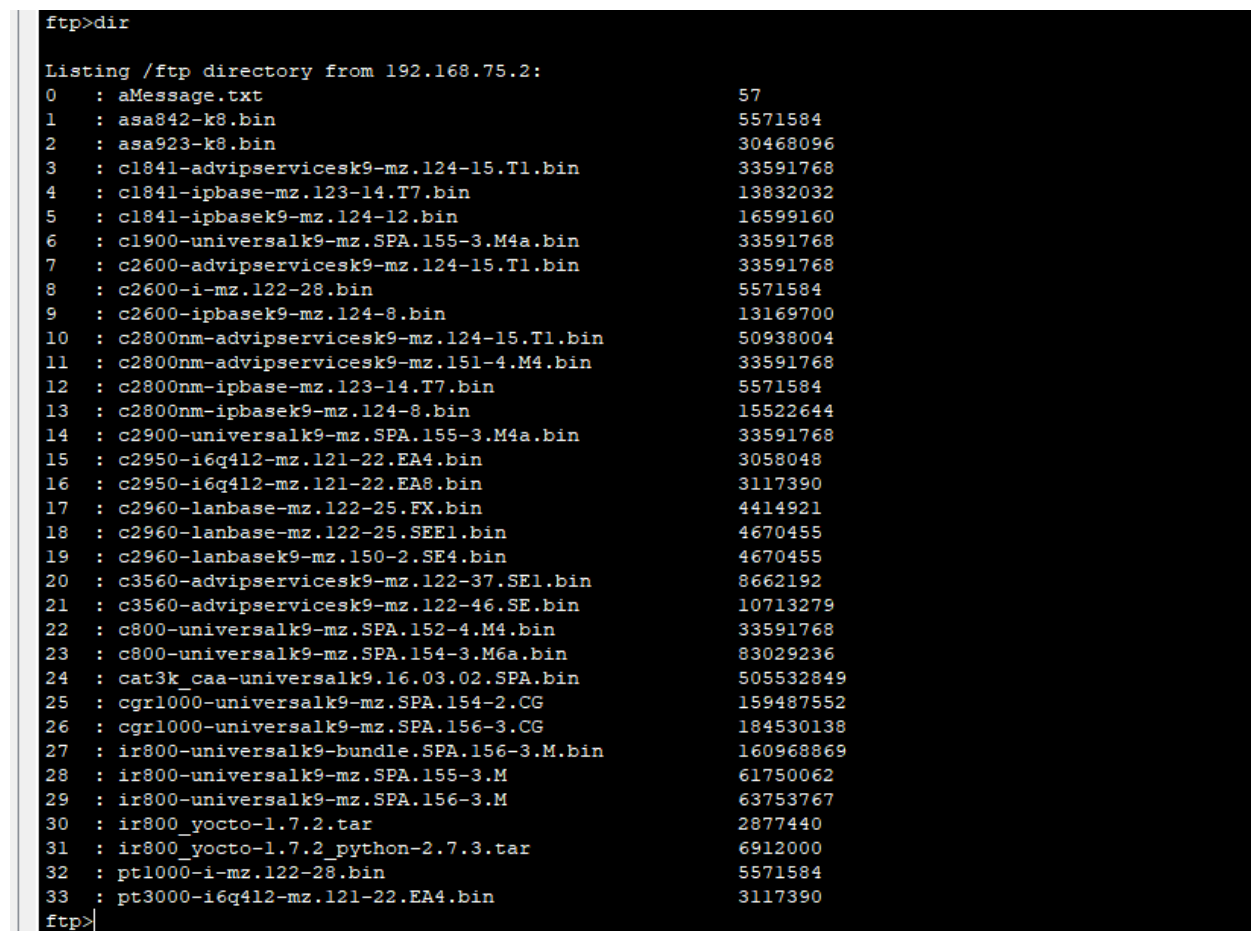
Username	Password	User Privilege
sukui	cisco123	RWDNL (Write, Read, Delete, Rename, List)
ajulo	cisco321	RWDNL (Write, Read, Delete, Rename, List)
malia	cisco123	RWNL (Write, Read, Rename, List)

Step 3: Transfer files between Net-Admin and the FTP server.

31. Click Net-Admin PC, and then click Desktop > Command Prompt.
32. Enter the command `ftp 192.168.75.2` to log in to the FTP server, and then authenticate with username `sukyi` and password `cisco123`.



33. Enter the `dir` command to list the files on the FTP server.



34. Use the get command to download aMessage.txt.

```
ftp>get aMessage.txt

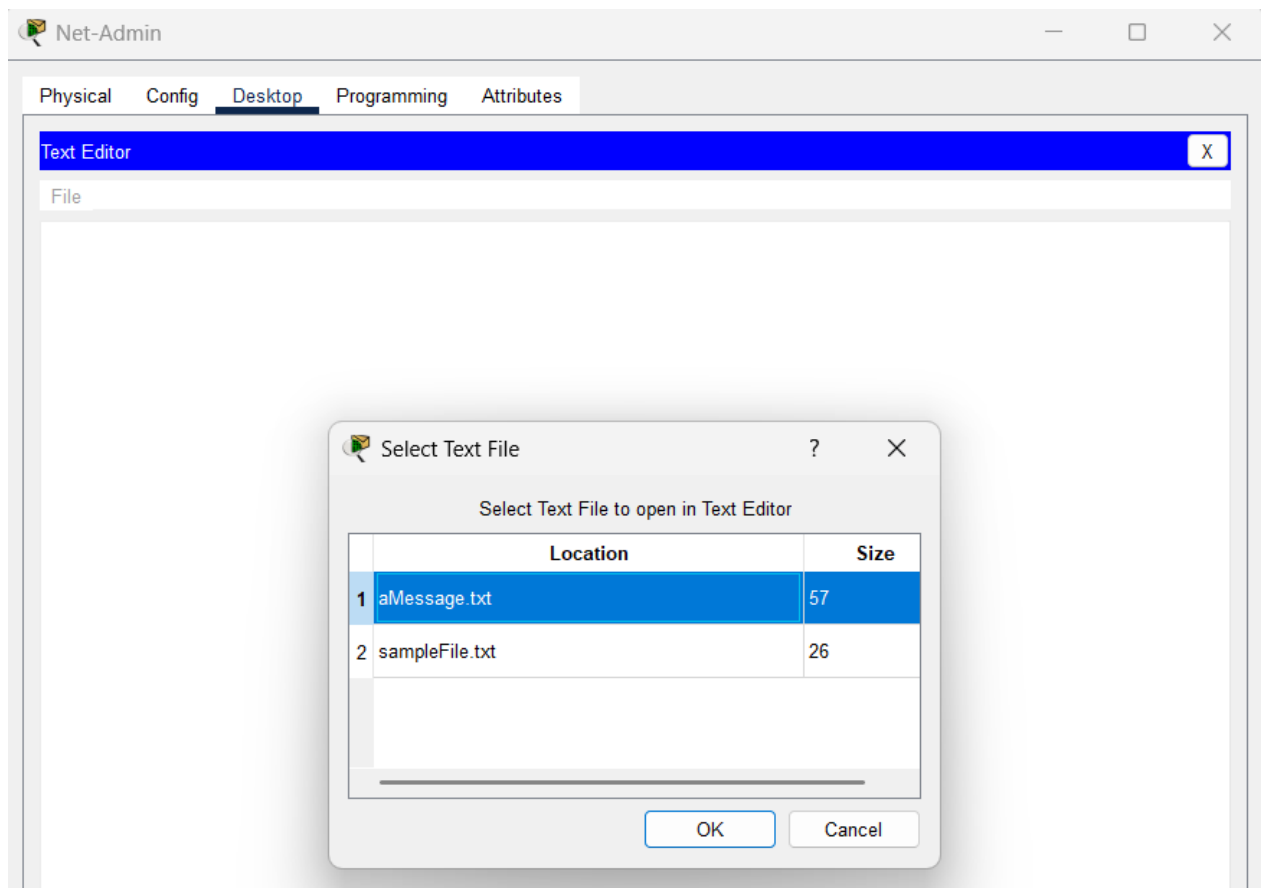
Reading file aMessage.txt from 192.168.75.2:
File transfer in progress...

[Transfer complete - 57 bytes]

57 bytes copied in 0 secs
ftp>
```

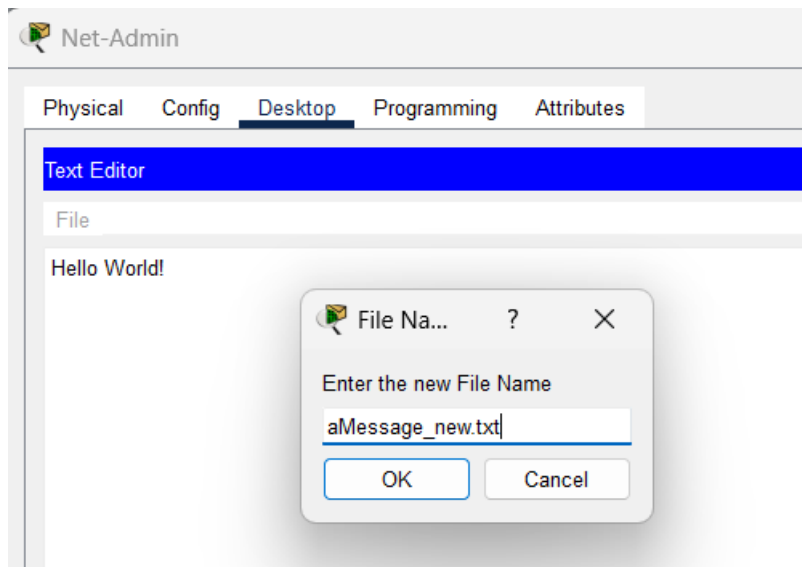
35. Quit the FTP session.

36. Close the Command Prompt, click Text Editor, and then File > Open. Open the downloaded file aMessage.txt.



37. Click File > New. Type a text message of your choice.

38. Click File > Save and save the new file as aMessage_new.txt. Close the Text Editor.



39. Click Command Prompt and then log back in to the FTP server as user sukyi.

40. Use the put command to upload aMessage_new.txt.

```
ftp>put aMessage_new.txt
Writing file aMessage_new.txt to 192.168.75.2:
File transfer in progress...

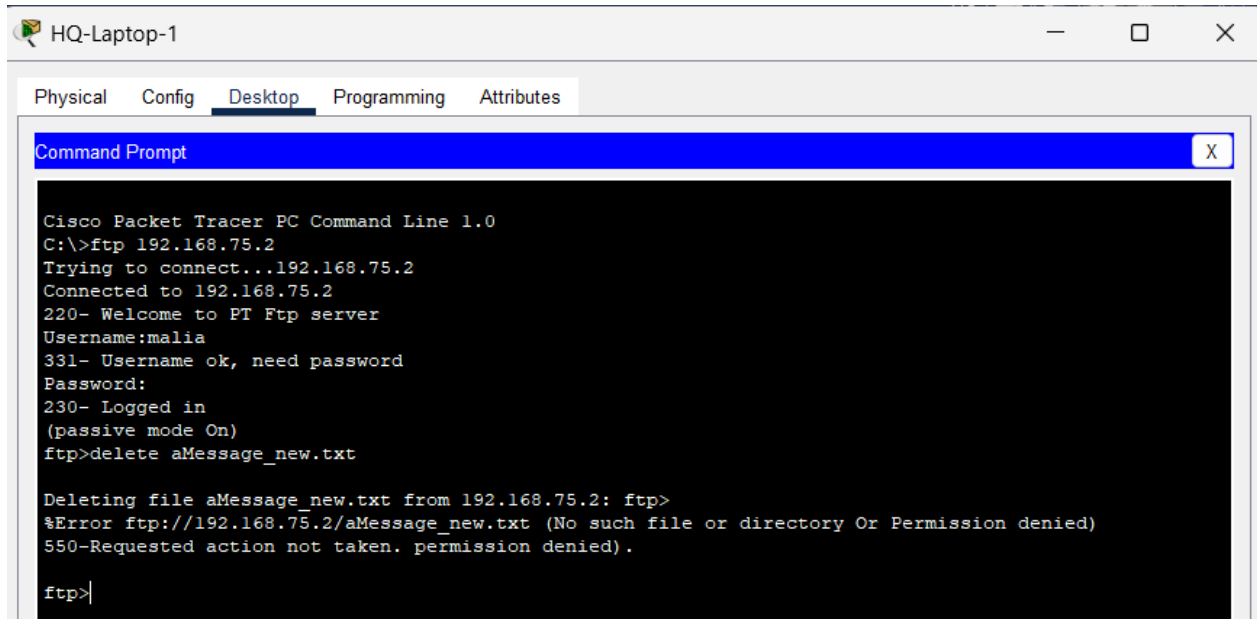
[Transfer complete - 12 bytes]

12 bytes copied in 0.057 secs (210 bytes/sec)
ftp>
```

41. Quit the FTP session.

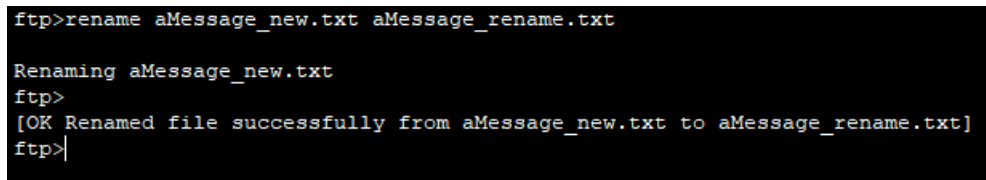
Step 4: Verify FTP user privileges are working as configured.

42. Navigate back to HQ and click HQ-Laptop-1, Desktop tab > Command Prompt.
43. Login to the FTP server at 192.168.75.2 with username malia and password cisco123.
44. Use the delete command to attempt to remove the newly uploaded file aMessage_new.txt.



- d. Use the rename command to attempt to change the name of aMessage_new.txt to aMessage_rename.txt.

```
ftp> rename aMessage_new.txt aMessage_rename.txt
```



45. Quit the FTP session and close the HQ-Laptop-1 window.